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Longitudinal associations among autism and ADHD traits, social maladjustment, and mental health in school-aged children

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Abstract

Background. Neurodevelopmental disorder traits are associated with a range of psychosocial difficulties, yet the longitudinal processes linking these traits to mental health remain insufficiently explored. This study examined developmental associations among autism spectrum disorder (ASD) traits, attention-deficit/hyperactivity disorder (ADHD) traits, social maladjustment, and mental health from childhood through adolescence.

Methods. Longitudinal data were drawn from 9,619 students across 11 grade cohorts in mainstream classrooms. Cross-lagged structural equation modeling was used to estimate independent longitudinal associations among ASD traits, ADHD traits, social maladjustment, and mental health, while controlling for confounding factors. ASD and ADHD traits were modeled simultaneously to clarify their unique developmental patterns.

Results. ASD traits were consistently associated with later internalizing problems and peer difficulties. ADHD traits were associated with later internalizing and externalizing problems, as well as academic challenges and strained teacher–parent relationships. These associations differed by trait type, suggesting distinct developmental pathways. Several bidirectional associations also emerged: poorer mental health was associated with subsequent social maladjustment, which was subsequently linked to persistent developmental difficulties.

Conclusions. The findings indicate that social maladjustment is involved in the longitudinal associations between developmental disorder traits and mental health and may shape developmental trajectories. These results highlight the need for comprehensive support approaches that consider peer relationships, learning environments, and family–school relationships.

Introduction

Autism spectrum disorder (ASD) and attention-deficit/hyperactivity disorder (ADHD) are neurodevelopmental disorders that exhibit high comorbidity rates with psychiatric disorders (Hossain et al., 2020; Larson, Russ, Kahn, & Halfon, 2011; Simonoff et al., 2008). Such comorbidities have also been observed in mild cases that do not meet the full diagnostic criteria of these disorders (Kim et al., 2009; Lundström et al., 2011; Rai et al., 2018), indicating that children with subthreshold symptoms often exhibit equivalent clinical support needs (Balázs & Keresztény, 2014; Kamio et al., 2013).

As mental-health–related problems during school age have long-term effects on educational, occupational, and social functioning in adulthood (Copeland, Wolke, Shanahan, & Costello, 2015), early intervention is a critical public health priority. However, the specific temporal pathways linking neurodevelopmental disorder traits to secondary mental health problems and the mechanisms for this association remain poorly understood. Moreover, minimal research has been conducted on children with subthreshold symptoms, and their support systems remain underdeveloped.

This study addresses this gap by using large-scale longitudinal data from Japanese elementary and middle-school students to examine longitudinal associations between ASD and ADHD traits, social adjustment, and mental health outcomes from childhood through adolescence. By treating neurodevelopmental characteristics as continuous spectra and examining their effects in a nonclinical population, this study aims to inform early intervention strategies and preventive support practices beyond traditional diagnostic frameworks.

Comorbidities of ASD/ADHD traits and mental health problems

A British study of children aged 10–14 years found that 70% of children with ASD had at least one additional diagnosis per the Diagnostic and Statistical Manual of Mental Disorders (DSM),

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whereas 40% had two or more comorbid conditions, with anxiety disorders and oppositional defiant disorder being particularly prevalent (Simonoff et al., 2008). Similarly, a large-scale epidemiological study in the United States examining children aged 6–17 years revealed that only 33.1% of children who had been diagnosed with ADHD presented without comorbidities or complications, with most exhibiting co-occurring conditions such as conduct disorder, anxiety disorders, and depression (Larson et al., 2011).

Comparable patterns have been observed for subthreshold ASD and ADHD traits, which are associated with elevated risks of anxiety, depressive symptoms, and externalizing problems (Kim et al., 2009; Lewinsohn, Shankman, Gau, & Klein, 2004; Lundström et al., 2011; Moriwaki & Kamio, 2013; Noda et al., 2013; Rai et al., 2018). These findings highlight the importance of examining mental health among children who present with elevated traits but do not meet diagnostic thresholds.

Processes linking ASD/ADHD traits and mental health

Although elevated ASD and ADHD traits are associated with increased mental health difficulties, the pathways underlying these associations remain unclear. One possible explanation for this link is that neurodevelopmental traits and psychiatric symptoms share biological underpinnings. For example, family and twin studies suggest genetic overlap between ADHD tendencies and emotional problems (Chen et al., 2016; Cole et al., 2009). However, genetic explanations alone may not fully account for the heterogeneity and high co-occurrence of symptoms, and evidence for shared biological mechanisms remains inconclusive.

Another possibility is that neurodevelopmental traits impede social adaptation, resulting in secondary mental health problems. For example, children with elevated ASD traits often experience challenges in peer interactions (Kasari, Locke, Gulsrud, & Rotheram-Fuller, 2011; Whitehouse, Durkin, Jaquet, & Ziatas, 2009). Similarly, children with elevated ADHD traits are prone to academic underachievement and interpersonal difficulties owing to concentration problems and impulsivity (Herman, Lambert, Ialongo, & Ostrander, 2007; Humphreys et al., 2013; Powell et al., 2020). Such adaptive difficulties may lead to decreased self-esteem, social isolation, and chronic stress, potentially resulting in secondary emotional problems. Nonetheless, causal pathways cannot be presumed on the basis of these associations alone, and alternative explanations, including moderation by contextual factors, remain plausible.

Social domains relevant to adjustment

Considering the practical importance of identifying school-based targets for support, this study focuses on three domains of social adjustment that have been consistently linked with ASD/ADHD traits and mental health outcomes.

Peer relationships

Children with ASD tend to have fewer friendships and lower friendship quality and often occupy peripheral positions in classroom social networks (Kasari et al., 2011; Whitehouse et al., 2009). Furthermore, population-based studies indicate that elevated ASD traits are linked with difficulties in maintaining friendships and increased risks of bullying victimization (Hsiao, Tseng, Huang, & Gau, 2013; Nakajima et al., 2021; Tanaka et al., 2014; Tanaka et al., 2015). Similarly, children with ADHD face heightened risks of peer rejection and bullying. Impulsivity and aggression contribute to exclusion, interpersonal conflict, and social isolation, with a substantial proportion of children with ADHD experiencing victimization regularly (Bagwell, Molina, Pelham, & Hoza, 2001; Becker, Mehari, Langberg, & Evans, 2017).

Relationships with teachers and parents

Children with ADHD are often reprimanded for impulsivity and noncompliance, which can strain parent–teacher relationships. These patterns are linked to increased parenting stress and harsher disciplinary practices, which may impair emotional regulation and increase internalization and externalization problems (Matsuoka et al., 2011; Rogers & Tannock, 2018). Conversely, children with ASD often show reduced attention to social cues and fewer attachment behaviors, making connections with supportive adults more difficult (Grzadzinski, Luyster, Spencer, & Lord, 2014; Jones, Carr, & Klin, 2008). Other research has highlighted that adult and peer support act as protective factors against depression, bullying, and school refusal (Jackson & Warren, 2000; Murayama et al., 2016; Williams, Connolly, Pepler, & Craig, 2005).

Academic problems

Children with ADHD often struggle with core academic skills such as reading, writing, and mathematics (Barkley, Anastopoulos, Guevremont, & Fletcher, 1991), with higher trait severity linked with poorer performance (DuPaul & Stoner, 2003; Kawabata, Tseng, & Gau, 2012). Classroom studies show greater off-task behavior and shorter attention spans among children with ADHD compared to their peers (Kofler, Rapport, & Alderson, 2008; Rapport et al., 2009). Moreover, large-scale research has confirmed higher rates of absenteeism, dropout, and poor outcomes among children with ADHD (Fleming et al., 2017). These findings highlight the broad academic impact of ADHD traits. Meanwhile, children with ASD may face difficulties in comprehension, limited interest in tasks, or avoidance of new content due to anxiety (Uno, 2020). Although some studies have reported discrepancies between intellectual ability and actual achievement in individuals with ASD without intellectual disability (Estes et al., 2011), evidence on the link between ASD traits and academic problems remains limited.

Study aim

While many previous studies have examined the associations between neurodevelopmental disorders and psychosocial problems, few have explored the specific social domains that may help explain how ASD and ADHD traits are associated with mental health outcomes. In particular, few studies have simultaneously modeled ASD and ADHD traits while longitudinally examining multiple domains of social adjustment (peer relationships, academic performance, and relationships with teachers and parents). To address this gap, this study analyzes six-wave longitudinal data from children in grades 4–9 to examine whether ASD and ADHD traits show longitudinal associations with internalizing (depression and anxiety) and externalizing problems (anger and aggression), and how these associations relate to difficulties in peer, academic, and adult–child relationship domains. The study does not test causal mediation but instead investigates temporal associations among these constructs.

Hypotheses

Social difficulties have been widely discussed as potential pathways linking neurodevelopmental traits and mental health. Longitudinal studies indicate that depressive symptoms in children with elevated ASD traits are associated with negative peer experiences, including

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bullying (Rai et al., 2018). Therefore, the following hypothesis was proposed:

Hypothesis 1: ASD traits are longitudinally associated with internalizing problems (depression and anxiety), and these associations are related to difficulties in peer relationships.

Academic challenges are particularly salient for ADHD. Children with elevated ADHD traits often experience persistent learning difficulties, which are associated with later emotional problems (Powell et al., 2020). Therefore, the following hypothesis was proposed:

Hypothesis 2: ADHD traits are longitudinally associated with internalizing problems (depression and anxiety), and these associations are related to academic difficulties.

ADHD traits have also been linked with externalizing problems in the context of interpersonal and environmental interactions. Reviews suggest that difficulties in emotion regulation contribute to aggression and oppositional behavior (Kraft, Ebner, Leo, & Lindenberg, 2023). Moreover, stressful or conflictual relationships with adults may exacerbate behavioral problems (Mak et al., 2020). These findings highlight the importance of relational contexts. Thus, the following hypothesis was proposed:

Hypothesis 3: ADHD traits are longitudinally associated with externalizing problems (anger and aggression), and these associations are related to difficulties in relationships with parents and teachers.

Methods

Participants and procedures

This study was conducted as part of an ongoing longitudinal cohort project. It aimed to elucidate the mechanisms underlying development and adaptation in children and adolescents through a comprehensive population-based survey encompassing all public elementary and junior high schools in a medium-sized city in Chubu, Japan.

The surveyed municipality is located within commuting distance of a major metropolitan area while maintaining robust industrial and agricultural sectors. Consequently, it encompasses households with diverse socioeconomic backgrounds, including families with members employed in urban centers and those engaged in locally based occupations. With a population of approximately 90,000, the municipality is nearly equivalent to the national average for cities and wards (approximately 100,000), excluding government-designated cities and Tokyo's 23 special wards.

Data were analyzed from longitudinal surveys conducted from FY2012 to FY2023. Information was collected from all enrolled students, their parents, and their homeroom teachers. To ensure the validity of the longitudinal analysis, students who participated in only one wave or none were excluded. The final sample included 9,619 students from an initial 11,549 enrolled students (participation rate: 83.3%). Details of cohort composition and grade progression are provided in Supplementary Table S1.

The data in this study comprise both complete longitudinal data from participants who continued across multiple waves and incomplete longitudinal data from those who participated in only some waves. Given the nature of school-based surveys, not all students could participate in every wave because of absences, school transfers, family circumstances, and other factors, leading to incomplete longitudinal data.

Comparisons of sex, student grade, and mean values of key variables showed no significant differences between the included and excluded students. All standardized mean differences were $d < .20$, suggesting that selection bias was limited. The average missing rate per wave was 11.6%, primarily due to absences and school transfers. This pattern is typical of school-based surveys, and the missingness was judged unlikely to violate the missing-at-random (MAR) assumption. To address the mixed presence of complete and incomplete longitudinal data, missing values were handled using full information maximum likelihood (FIML), using all available information for estimation.

Before data collection, informed consent was obtained from the students and parents via explanatory documents stating that participation was voluntary and that no disadvantages would result from nonparticipation. This study was reviewed and approved by our institution's Ethics Committee.

Survey content

Developmental characteristics

ASD and ADHD tendencies were measured using parent-report measures: the shortened Japanese versions of the Autism Spectrum Screening Questionnaire (ASSQ; Ito, Matsumoto, et al., 2014a; Ito, Tanaka, et al., 2014b) and the ADHD Rating Scale-IV (DuPaul, Power, Anastopoulos, & Reid, 1998; Ichikawa & Tanaka, 2008; Tanaka et al., 2015). The ASSQ primarily captures Asperger-like social and behavioral characteristics, which represent a narrower subset of the broader ASD phenotype described in DSM-5 and ICD-11. Both scales are suitable for assessing dimensional variation in neurodevelopmental traits within nonclinical school populations.

Social maladjustment

Social maladjustment was measured using the Social Maladjustment Scale for Elementary and Junior High School Students (Ito, Matsumoto, et al., 2014a; Ito, Tanaka, et al., 2014b), a self-report questionnaire that captures children's subjective sense of maladjustment. The scale evaluates difficulties in three domains: academic problems (e.g. 'Has difficulty keeping up with classwork'), problems with teachers (e.g. 'Often conflicts with teachers'), and family problems (e.g. 'Frequent arguments with parents').

Peer problems were assessed using the five-item 'peer relationships' subscale from the Japanese version of the Strengths and Difficulties Questionnaire (SDQ; Harada et al., 2014). The ratings obtained from students, teachers, and parents were z-standardized and then combined into a composite variable ($\alpha = .898$). The multi-informant SDQ approach reduces rater bias and captures peer difficulties across home and school contexts. The SDQ Peer Problems subscale is internationally validated and complements aspects of social functioning not fully represented in the Social Maladjustment Scale.

Mental health

Depression was measured using the shortened Japanese Birleson Depression Self-Rating Scale for Children (DSRS-C; Murata, Shimizu, Mori, & Oshima, 1996; Namikawa et al., 2011). Aggression was assessed using the shortened Hostility-Aggression Questionnaire for Children (HAQ-C; Sakai et al., 2000). Teacher and parent SDQ ratings of emotional symptoms and conduct problems were integrated with DSRS-C and HAQ-C to form composite variables for internalizing ($\alpha = .861$) and externalizing problems ($\alpha = .895$).

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Statistical analysis

We used structural equation modeling (SEM) to estimate cross-lagged panel models across six waves. A conceptual diagram of the longitudinal associations examined in this study is presented in Supplementary Figure S1.

Analyses were conducted using JASP (Version 0.18.3; JASP Team, 2024). Parameters were estimated with robust maximum likelihood (MLR), which provides standard errors and test statistics that are robust to non-normality. Missing data were handled using FIML, incorporating all available data under the MAR assumption.

The cross-lagged panel model included autoregressive paths for each construct across adjacent waves; cross-lagged paths among ASD and ADHD traits; social maladjustment variables (peer, academic, teacher, and family problems); internalizing and externalizing problems; and within-wave correlations among all variables. All primary variables were *z*-standardized to facilitate comparison of standardized coefficients. Because the model was saturated (df = 0), fit indices were not informative and therefore were not reported. To reduce potential confounding, sex and school grade were included as time-invariant covariates in all models.

Results

Descriptive statistics and correlation coefficients

Summary statistics, distributional characteristics, reliability coefficients, and correlations for all major variables are provided in Supplementary Tables S2–S4 and Supplementary Figure S2. Overall, the variables showed right-skewed distributions with most participants scoring in the lower range.

Tables 1–3 present descriptive statistics and correlation coefficients for each fourth- and ninth-grade variable. ASD traits showed a slight decrease from fourth to ninth grade, whereas ADHD traits demonstrated a more pronounced decline. In contrast, academic problems and depressive symptoms increased across development. ASD and ADHD traits were moderately correlated at both time points, and ASD traits showed strong associations with peer problems. Peer problems were also strongly associated with internalizing symptoms.

Cross-lagged effects

Cross-lagged structural equation models were used to examine longitudinal associations among variables. Regression coefficients were interpreted using Orth et al.'s (2024) guidelines (~.03 = small, .07 = medium, .12 = large). All effects reported below reflect temporal associations rather than causal influence.

The major cross-lagged effects are shown in Figure 1, and the full set of cross-lagged coefficients is summarized in Table 4.

Internalizing problems

ASD traits showed small-to-medium longitudinal associations with later internalizing problems (β = .068–.097). Associations were relatively stronger from fourth to fifth grade (β = .093) and seventh to eighth grade (β = .097). Peer problems also showed small-to-large associations with later internalizing problems (β = .046–.121), with the largest association observed from fifth to sixth grade (β = .121). Academic problems during elementary school showed small associations with later internalizing problems (β = .055–.090). These findings indicate that ASD traits, peer difficulties, and academic

Table 1. Descriptive statistics for each variable

		4th grade		5th grade		6th grade		7th grade		8th grade		9th grade	
	Informant	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*	*M*	*SD*
ASD traits	Parent	13.14	(2.94)	13.13	(2.94)	13.09	(2.92)	13.03	(2.91)	12.97	(2.83)	12.90	(2.76)
ADHD traits	Parent	2.93	(3.71)	2.59	(3.53)	2.14	(3.31)	2.17	(3.23)	2.01	(3.06)	1.69	(2.79)
Social maladaptation													
Peer problems													
Peer problems	Child	2.68	(1.68)	2.38	(1.63)	2.17	(1.57)	2.05	(1.54)	2.00	(1.54)	2.02	(1.55)
Peer problems	Teacher	6.05	(1.54)	6.06	(1.56)	5.87	(1.42)	6.14	(1.59)	6.22	(1.70)	6.22	(1.68)
Peer problems	Parent	1.60	(1.63)	1.57	(1.61)	1.53	(1.60)	1.55	(1.61)	1.57	(1.62)	1.58	(1.64)
Teacher problems	Child	0.98	(1.42)	0.86	(1.37)	0.79	(1.35)	0.69	(1.29)	0.91	(1.52)	0.77	(1.35)
Family problems	Child	2.03	(1.70)	1.79	(1.67)	1.65	(1.69)	1.69	(1.80)	1.71	(1.85)	1.59	(1.82)
Academic problems	Child	3.40	(2.06)	3.41	(2.09)	3.30	(2.08)	4.15	(2.10)	4.51	(2.16)	4.45	(2.14)
Internalizing problems													
Depression	Child	4.22	(2.88)	4.27	(2.93)	4.32	(3.06)	4.62	(3.19)	4.97	(3.34)	5.19	(3.39)
Emotional symptoms	Teacher	5.84	(1.51)	5.76	(1.46)	5.60	(1.24)	5.68	(1.43)	5.83	(1.59)	5.84	(1.71)
Emotional symptoms	Parent	1.66	(1.85)	1.60	(1.84)	1.44	(1.75)	1.42	(1.76)	1.39	(1.75)	1.41	(1.79)
Externalizing problems													
Aggression	Child	16.00	(5.61)	16.14	(5.39)	16.00	(5.24)	15.51	(5.13)	15.21	(4.93)	14.70	(4.84)
Conduct problems	Teacher	6.05	(1.62)	6.01	(1.62)	5.79	(1.37)	5.80	(1.40)	5.85	(1.48)	5.80	(1.43)
Conduct problems	Parent	1.76	(1.52)	1.70	(1.51)	1.58	(1.44)	1.57	(1.42)	1.55	(1.43)	1.43	(1.36)

ADHD, attention-deficit/hyperactivity disorder; ASD, autism spectrum disorder; SDQ, Strengths and Difficulties Questionnaire.

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Table 2. Correlation coefficients among variables at fourth grade

		1	2	3	4	5	6	7	8
1	ASD traits	–							
2	ADHD traits	.559***	–						
3	Peer problems	.550***	.348***	–					
4	Teacher problems	.159***	.229***	.243***	–				
5	Family problems	.166***	.195***	.240***	.430***	–			
6	Academic problems	.128***	.222***	.227***	.364***	.389***	–		
7	Internalizing problems	.399***	.286***	.541***	.201***	.271***	.269***	–	
8	Externalizing problems	.329***	.452***	.388***	.395***	.341***	.317***	.315***	–

Note: *p < .05 **p < .01 ***p < .001.

Table 3. Correlation coefficients among variables at ninth grade

		1	2	3	4	5	6	7	8
1	ASD traits	–							
2	ADHD traits	.518***	–						
3	Peer problems	.573***	.310***	–					
4	Teacher problems	.094***	.146***	.173***	–				
5	Family problems	.155***	.166***	.223***	.387***	–			
6	Academic problems	.076***	.167***	.110***	.258***	.220***	–		
7	Internalizing problems	.375***	.270***	.597***	.199***	.307***	.211***	–	
8	Externalizing problems	.270***	.380***	.370***	.304***	.318***	.227***	.362***	–

Note: *p < .05 **p < .01 ***p < .001.

challenges each show temporal associations with internalizing symptoms across development.

Externalizing problems

ADHD traits showed small-to-medium longitudinal associations with later externalizing problems (β = .064–.118). Associations were strongest from fifth to sixth grade (β = .118) and fourth to fifth grade (β = .107). Family problems also showed small-to-medium associations with later externalizing problems (β = .029–.093). These patterns suggest that ADHD traits and family-related difficulties are consistently associated with externalizing behaviors over time.

Peer problems

ASD traits showed large longitudinal associations with later peer problems (β = .146–.190). Associations were consistently strong across all grade transitions. Internalizing problems also showed small-to-medium associations with later peer problems (β = .043–.107). These results indicate stable bidirectional associations between ASD traits, emotional difficulties, and peer relationship challenges.

Teacher problems

Externalizing problems showed medium-to-large associations with later teacher relationship problems (β = .103–.262), with the strongest association from seventh to eighth grade (β = .262). Academic problems also showed medium-to-large associations with later teacher problems (β = .082–.128). These findings suggest that behavioral and academic difficulties are consistently associated with strained teacher relationships.

Family problems

Academic problems showed medium-to-large associations with later family problems (β = .105–.163). Externalizing problems also showed small-to-medium associations with later family problems (β = .042–.126). These results indicate that academic and behavioral difficulties are associated with increased family-related challenges over time.

Academic problems

ADHD traits showed small-to-medium associations with later academic problems (β = .015–.069). Family problems also showed medium-to-large associations with later academic problems (β = .090–.161). These findings highlight consistent associations between ADHD traits, family context, and academic functioning.

ASD traits

Peer problems showed medium-to-large associations with later ASD traits (β = .096–.164), with the strongest associations from sixth to seventh grade (β = .164) and eighth to ninth grade (β = .145). These results suggest reciprocal associations between ASD traits and peer difficulties across development.

ADHD traits

Externalizing problems showed small-to-medium associations with later ADHD traits (β = .032–.082). Associations were strongest from seventh to eighth grade (β = .082). These findings indicate reciprocal associations between ADHD traits and behavioral difficulties.

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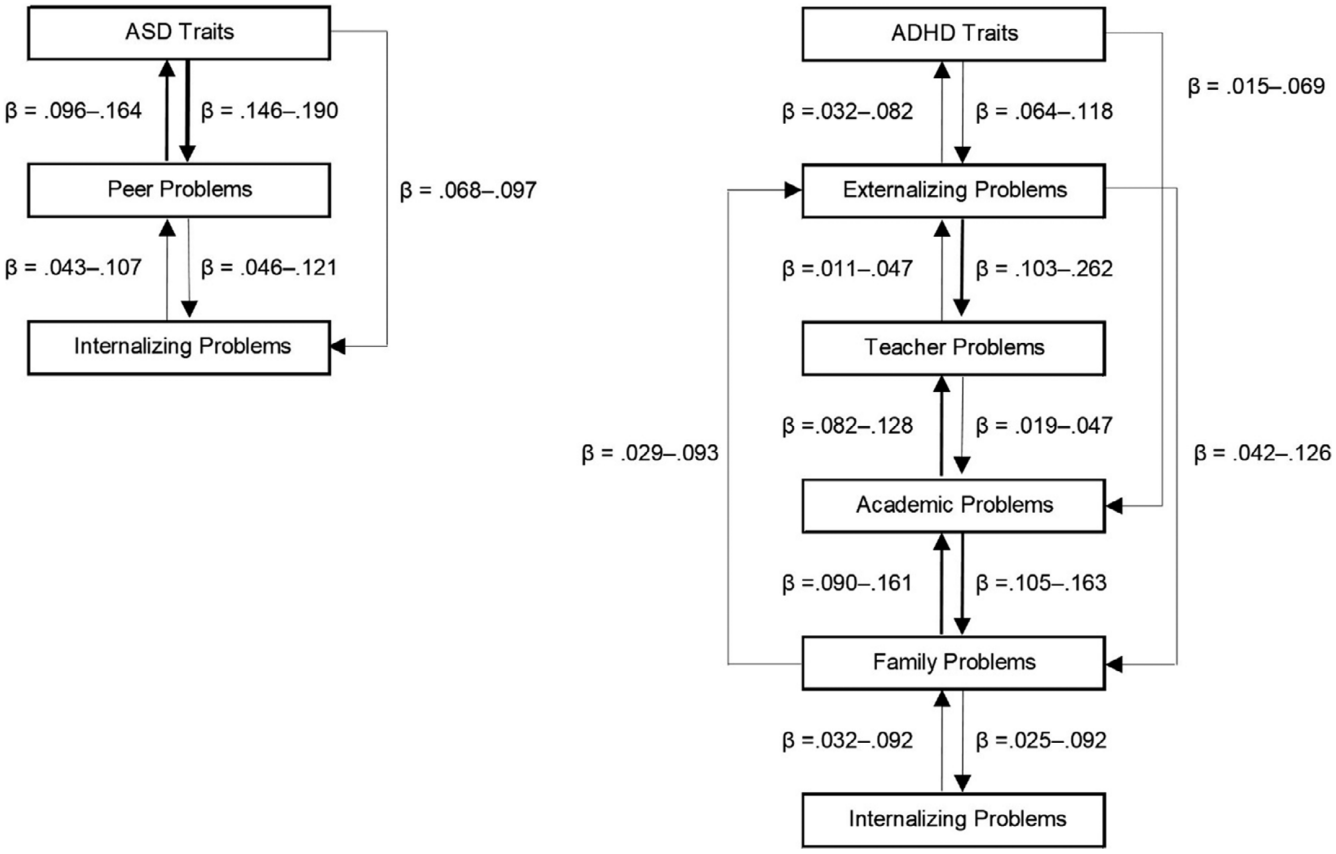


Figure 1. Cross-lagged path models for ASD traits and ADHD traits.

Discussion

This study examined longitudinal associations among ASD traits, ADHD traits, social adjustment, and mental health outcomes in a large cohort of Japanese children and adolescents. The findings highlight distinct patterns of association for ASD and ADHD traits across multiple social domains and underscore the importance of considering dynamic, bidirectional processes rather than unidirectional causal pathways.

Association between ASD traits and internalizing problems

The results showed that ASD traits were associated with peer relationship difficulties over time. Moreover, peer relationship difficulties were subsequently associated with increases in internalizing symptoms, supporting Hypothesis 1. However, these associations do not establish mediation or causal direction. This pattern aligns with prior research reporting that peer problems are related to emotional difficulties among youth with elevated ASD traits (Rai et al., 2018).

These associations were particularly pronounced during the upper elementary years, a developmental period characterized by increasing importance of peer group belonging and social competence. Difficulties in social communication and peer interactions associated with ASD traits may place children at heightened risk for emotional distress, especially as peer relationships become more central to daily life. These findings emphasize the importance of school-based support systems that strengthen peer relationships, such as structured social skills programs and peer-mediated interventions, which may help reduce emotional distress during this sensitive developmental stage.

Association between ADHD traits and internalization

Although ADHD traits were significantly associated with internalizing problems at several grade intervals, the magnitude of these associations was generally small, suggesting limited direct links. Instead, ADHD traits were more consistently associated with academic difficulties, which themselves were associated with later internalizing symptoms. This pattern is consistent with prior work showing that inattention and impulsivity are linked with academic challenges (Barkley et al., 1991; DuPaul & Stoner, 2003; Kawabata et al., 2012).

The results provide partial support for Hypothesis 2: academic difficulties may represent one pathway through which ADHD traits relate to internalizing symptoms. However, these associations were not consistent across all developmental periods and do not indicate formal mediation. These findings suggest that academic support – such as individualized instruction, organizational skills training, and self-efficacy enhancement – may help reduce emotional distress among children with elevated ADHD traits.

Association between ADHD traits and externalizing problems

ADHD traits showed strong and consistent longitudinal associations with externalizing problems, particularly during elementary school. This pattern is consistent with the core behavioral characteristics of ADHD, including impulsivity and difficulties with behavioral regulation, which are often linked with oppositional and aggressive behaviors.

Difficulties in relationships with teachers and parents were also associated with externalizing problems, although these associations

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Table 4. Cross-lagged effects of each variable

		4th_5th		5th_6th		6th_7th		7th_8th		8th_9th	
Dependent variable	Independent variable	*β*	*SE*	*β*	*SE*	*β*	*SE*	*β*	*SE*	*β*	*SE*
Internalizing problems	ASD traits	.093***	.016	.079***	.014	.068***	.015	.097***	.015	.071***	.015
	ADHD traits	.053***	.015	.026*	.013	.007	.014	−.020	.014	.043**	.014
	Peer problems	.058***	.022	.121***	.019	.092***	.019	.082***	.020	.046**	.020
	Teacher problems	−.033*	.014	−.001	.012	−.006	.012	−.027*	.012	.046***	.012
	Family problems	.025	.014	.051***	.012	.076***	.012	.091***	.012	.092***	.012
	Academic problems	.090***	.013	.055***	.011	.055***	.012	.021	.011	−.006	.011
	Internalizing problems	.490***	.021	.503***	.019	.495***	.019	.509***	.020	.545***	.021
	Externalizing problems	−.051***	.021	−.057***	.019	−.005	.019	−.013	.019	−.065***	.019
Externalizing problems	ASD traits	.038*	.015	.029*	.013	.021	.014	.036*	.014	.056***	.015
	ADHD traits	.107***	.014	.118***	.012	.089***	.013	.064***	.013	.078***	.014
	Peer problems	.037*	.020	.029*	.018	.012	.018	.000	.019	−.003	.020
	Teacher problems	.013	.013	.016	.011	.022*	.011	.023*	.011	.047***	.012
	Family problems	.029*	.013	.042***	.011	.074***	.011	.068***	.012	.093***	.012
	Academic problems	.062***	.012	.049***	.010	.026*	.011	.029**	.011	.006	.011
	Internalizing problems	−.050***	.020	−.066***	.017	−.030*	.018	−.030*	.019	−.039**	.021
	Externalizing problems	.580***	.019	.610***	.017	.600***	.018	.611***	.018	.545***	.019
Peer problems	ASD traits	.179***	.015	.190***	.014	.146***	.015	.167***	.015	.168***	.014
	ADHD traits	.015	.014	.007	.013	−.010	.014	−.017	.014	.012	.013
	Peer problems	.485***	.021	.530***	.019	.530***	.019	.708***	.020	.570***	.019
	Teacher problems	.019	.013	.005	.012	.005	.012	−.002	.012	.021	.011
	Family problems	.007	.013	.024*	.012	.051***	.012	.054***	.012	.064***	.012
	Academic problems	.029*	.013	.014	.011	−.006	.012	−.013	.011	−.043***	.011
	Internalizing problems	.069***	.020	.056***	.019	.075***	.019	.107***	.020	.043**	.020
	Externalizing problems	.020	.020	.002	.019	.033*	.019	.013	.019	−.014	.018
Teacher problems	ASD traits	−.010	.018	−.006	.017	−.006	.017	−.041*	.017	.018	.016
	ADHD traits	.093***	.017	.013	.016	.024	.016	−.006	.015	−.008	.016
	Peer problems	.022	.025	.018	.023	−.013	.022	−.014	.023	.003	.023
	Teacher problems	.225***	.016	.230***	.014	.238***	.014	.263***	.013	.307***	.013
	Family problems	.065***	.016	.079***	.014	.083***	.014	.055***	.014	.117***	.014
	Academic problems	.106***	.015	.104***	.013	.110***	.013	.128***	.013	.082***	.013
	Internalizing problems	−.034*	.024	−.012	.022	.000	.022	.003	.023	.028	.023
	Externalizing problems	.107***	.024	.125***	.022	.121***	.021	.262***	.021	.103***	.021
Family problems	ASD traits	.030	.018	.015	.016	.011	.016	.011	.016	.022	.015
	ADHD traits	.034*	.017	−.014	.015	.010	.015	.002	.015	−.010	.015
	Peer problems	.013	.024	.024	.021	.014	.021	−.003	.022	.011	.021
	Teacher problems	.024	.015	−.010	.013	.051***	.013	.042***	.013	.034**	.012
	Family problems	.284***	.015	.370***	.013	.384***	.013	.384***	.013	.419***	.013
	Academic problems	.163***	.015	.133***	.012	.105***	.012	.156***	.012	.160***	.012
	Internalizing problems	.065***	.023	.032*	.021	.050***	.021	.073***	.022	.092***	.022
	Externalizing problems	.071***	.023	.104***	.021	.079***	.02	.126***	.020	.042**	.020
Academic problems	ASD traits	−.050**	.018	−.035*	.016	−.028	.016	−.037*	.016	−.020	.016
	ADHD traits	.069***	.017	.058***	.015	.015	.015	.027	.015	.027	.015

(*Continued*)

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Table 4. (*Continued*)

Dependent variable	Independent variable	4th_5th		5th_6th		6th_7th		7th_8th		8th_9th	
		β	*SE*	*β*	*SE*	*β*	*SE*	*β*	*SE*	*β*	*SE*
	Peer problems	.037*	.024	.009	.021	.018	.021	−.055*	.022	.005	.021
	Teacher problems	.035*	.015	.019	.013	.036**	.013	.019	.013	.047***	.013
	Family problems	.090***	.015	.098***	.013	.102***	.013	.129***	.013	.161***	.013
	Academic problems	.365***	.015	.419***	.012	.402***	.013	.436***	.012	.418***	.012
	Internalizing problems	.048**	.023	.053***	.021	.042**	.021	.033	.022	.057***	.022
	Externalizing problems	.016	.023	.015	.021	.052***	.020	.106***	.020	.029*	.020
ASD traits	ASD traits	.685***	.012	.648***	.012	.600***	.012	.630***	.012	.667***	.012
	ADHD traits	.069***	.012	.120***	.011	.079***	.012	.076***	.011	.076***	.011
	Peer problems	.096***	.017	.102***	.016	.164***	.016	.127***	.017	.145***	.016
	Teacher problems	.000	.011	−93.9	.010	−.012	.010	−.028**	.010	.029**	.009
	Family problems	−.002	.011	.018	.010	.008	.010	.016	.010	.004	.010
	Academic problems	.019	.010	−.010	.009	−97.3	.010	−.003	.009	.006	.009
	Internalizing problems	−.009	.017	.008	.015	.013	.016	.029**	.017	.000	.016
	Externalizing problems	.005	.016	−.012	.015	.002	.016	.003	.015	−.037***	.015
ADHD traits	ASD traits	.119***	.013	.071***	.012	.066***	.013	.125***	.013	.107***	.013
	ADHD traits	.656***	.013	.680***	.011	.645***	.012	.604***	.012	.626***	.013
	Peer problems	.019	.018	.024*	.016	.022	.017	−.027*	.018	.015	.018
	Teacher problems	.005	.011	.016	.010	.013	.010	.019	.010	.019	.011
	Family problems	−.011	.011	−.009	.010	.011	.011	.026*	.011	.011	.011
	Academic problems	.036***	.011	.015	.009	.031**	.010	.012	.010	.034***	.010
	Internalizing problems	−.016	.017	−.013	.016	.006	.017	.030*	.018	−.005	.019
	Externalizing problems	.052***	.017	.060***	.016	.058***	.016	.082***	.016	.032**	.017

Note: *$p < .05$ **$p < .01$ ***$p < .001$.

were weaker than those involving ADHD traits, providing only partial support for Hypothesis 3. These findings highlight the importance of early behavioral interventions targeting self-regulation, combined with collaborative support between families and schools, to help reduce the escalation of behavioral difficulties.

Bidirectional associations between social maladjustment and mental health

A key contribution of this study is the identification of bidirectional associations between social adjustment and mental health. Peer difficulties and internalizing problems showed mutual associations, as did teacher–student and parent–child relationship problems with externalizing behaviors. These findings suggest that social difficulties and mental health symptoms may reinforce one another over time, forming dynamic cycles rather than simple one-way pathways.

For example, peer isolation may be associated with increases in depressive symptoms, while emotional distress may, in turn, be associated with further withdrawal or victimization. Similarly, strained relationships with teachers or parents may be associated with increases in oppositional behavior, which may contribute to additional relational conflict. These patterns underscore the importance of school-based mental health services, including counseling and early identification systems, to interrupt these cycles and provide continuous support throughout adolescence.

Interaction between developmental traits and social environment

The findings also indicate that ASD and ADHD traits are not static characteristics; rather, their subsequent levels may be associated with ongoing interactions with social environments. Peer difficulties were associated with later increases in ASD traits, and externalizing problems were associated with later increases in ADHD traits, suggesting reciprocal associations between traits and social experiences. These results align with the International Classification of Functioning, Disability and Health (ICF) framework, which conceptualizes functioning as an interaction between individual characteristics and environmental contexts.

This perspective is also consistent with Bronfenbrenner's (1979) ecological systems theory, in which peer relationships, teacher–student interactions, and academic contexts represent key microsystems that shape developmental outcomes. Understanding ASD and ADHD traits within these dynamic systems highlights the importance of interventions that address both individual needs and environmental supports, including classroom accommodations, teacher training, and family–school collaboration.

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Limitations

Several limitations should be acknowledged. First, clinical diagnostic information, including ASD, ADHD, and internalizing/externalizing disorders, was unavailable, and the study relied on dimensional trait measures. Given the size of the sample, it is likely that some children with clinical diagnoses were included, which may have influenced the distribution or variability of the measured constructs. However, because the analyses focused on dimensional traits rather than diagnostic groups, the inclusion of clinically diagnosed cases does not undermine the validity of the dimensional approach. Second, although six waves of data were analyzed, the follow-up intervals varied and the design did not constitute a fully closed cohort, which limits causal interpretation despite the prospective ordering of variables.

Third, the high correlations observed among some constructs, particularly between ASD traits and internalizing problems, may reflect item overlap, shared method variance, or rater biases. Future studies should incorporate multiple measurement approaches, including observational methods and performance-based assessments, to reduce shared-method dependencies. Fourth, the distribution of ASD traits deviated from normality, showing the positively skewed pattern typically observed in general-population samples. This non-normality may have influenced parameter estimates, including regression coefficients and variance estimation.

Fifth, formal tests of longitudinal measurement invariance across waves could not be conducted because identical item-level data were not available across all waves. This raises the possibility that changes in psychometric properties may have affected longitudinal estimates. Sixth, some variables were assessed through self-report, which may be influenced by subjective perceptions and response styles. In addition, the composite variables used in the analyses required integrating information across informants; although the construction procedures were described in detail, the weighting scheme may still introduce measurement heterogeneity.

Seventh, although this study examined prospective associations, it did not formally test mediation or indirect effects. The social adaptation indicators included in the model were treated as correlates rather than mediators, and the study did not evaluate whether these variables moderated the associations between developmental traits and mental health outcomes. Thus, the possibility that social contexts may buffer or amplify these associations remains an important direction for future research.

Eighth, assumptions related to missing data were based on the MAR framework, but MAR could not be empirically verified. Similarly, although standardized mean differences for attrition analyses were generally below 0.20, indicating limited selection bias, these thresholds should be interpreted cautiously. Finally, the sample was drawn from mainstream classrooms in a single region of Japan, which may limit the generalizability of the findings to other cultural or educational contexts.

Conclusions

This study examined longitudinal associations among ASD traits, ADHD traits, social adjustment, and mental health outcomes using large-scale cohort data. The findings showed that social environmental maladjustment was not a simple mediating mechanism but was a part of a broader set of temporal associations linking developmental traits and mental health. In particular, bidirectional associations between social adjustment and mental health suggest that these processes unfold dynamically over time rather than through a single, unidirectional pathway.

Furthermore, the results indicate that social contexts, such as peer relationships, academic environments, and interactions with teachers and parents, play a meaningful role in shaping developmental trajectories. These contexts were associated with both later mental health outcomes and subsequent levels of ASD and ADHD traits, underscoring the importance of understanding developmental characteristics within their environmental settings.

Overall, these findings highlight the need for comprehensive, context-sensitive support strategies that address both individual characteristics and the social environments in which children develop. Interventions that strengthen peer relationships, enhance academic support, and improve family–school collaboration may help promote healthier developmental pathways. By emphasizing dynamic interactions between traits and environments, this study provides a conceptual foundation for developing integrated support systems that respond to both social and individual needs.

Supplementary material. The supplementary material for this article can be found at http://doi.org/10.1017/S003329172610539X.

Data availability statement. The data that support the findings of this study are available from the corresponding author, EH, upon reasonable request.

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Competing interests. The authors declare none.

Ethical standards. The authors assert that all procedures contributing to this work comply with the ethical standards of the relevant national and institutional committees on human experimentation and with the Helsinki Declaration of 1975, as revised in 2008. This study was reviewed and approved by the Ethics Committee of Chukyo University (approved March 2007, reference number 18–166).

Informed consent. Informed consent was obtained from the parents of all participants before their inclusion in the study.

References

Bagwell, C. L., Molina, B. S., Pelham, W. E., & Hoza, B. (2001). Attention-deficit hyperactivity disorder and problems in peer relations: Predictions from childhood to adolescence. *Journal of the American Academy of Child and Adolescent Psychiatry*, **40**(11), 1285–1292. https://doi.org/10.1097/00004583-200111000-00008

Balázs, J., & Keresztény, Á. (2014). Subthreshold attention deficit hyperactivity in children and adolescents: A systematic review. *European Child and Adolescent Psychiatry*, **23**(6), 393–408. https://doi.org/10.1007/s00787-013-0514-7

Barkley, R. A., Anastopoulos, A. D., Guevremont, D. C., & Fletcher, K. E. (1991). Adolescents with ADHD: Patterns of behavioral adjustment, academic functioning, and treatment utilization. *Journal of the American Academy of Child & Adolescent Psychiatry*, **30**(5), 752–761. https://doi.org/10.1016/s0890-8567(10)80010-3

Becker, S. P., Mehari, K. R., Langberg, J. M., & Evans, S. W. (2017). Rates of peer victimization in young adolescents with ADHD and associations with

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internalizing symptoms and self-esteem. *European Child and Adolescent Psychiatry*, **26**(2), 201–214. https://doi.org/10.1007/s00787-016-0881-y

Bronfenbrenner, U. (1979). *The ecology of human development: Experiments by nature and design*. Harvard University Press.

Chen, T. J., Ji, C. Y., Wang, S. S., Lichtenstein, P., Larsson, H., & Chang, Z. (2016). Genetic and environmental influences on the relationship between ADHD symptoms and internalizing problems: A Chinese twin study. *American Journal of Medical Genetics Part B: Neuropsychiatric Genetics*, **171**(7), 931–937. https://doi.org/10.1002/ajmg.b.32411

Cole, J., Ball, H. A., Martin, N. C., Scourfield, J., & McGuffin, P. (2009). Genetic overlap between measures of hyperactivity/inattention and mood in children and adolescents. *Journal of the American Academy of Child & Adolescent Psychiatry*, **48**(11), 1094–1101. https://doi.org/10.1097/CHI.0b013e3181b7666e

Copeland, W. E., Wolke, D., Shanahan, L., & Costello, E. J. (2015). Adult functional outcomes of common childhood psychiatric problems: A prospective, longitudinal study. *JAMA Psychiatry*, **72**(9), 892–899. https://doi.org/10.1001/jamapsychiatry.2015.0730

DuPaul, G. J., Power, T. J., Anastopoulos, A., & Reid, R. (1998). *ADHD rating scale-IV: Checklists, norms, and clinical interpretation*. Guilford Press.

DuPaul, G. J., & Stoner, G. (2003). *ADHD in the schools: Assessment and intervention strategies* (2nd ed.). Guilford Press.

Estes, A., Rivera, V., Bryan, M., Cali, P., & Dawson, G. (2011). Discrepancies between academic achievement and intellectual ability in higher-functioning school-aged children with autism spectrum disorder. *Journal of Autism and Developmental Disorders*, **41**(8), 1044–1052. https://doi.org/10.1007/s10803-010-1127-3

Fleming, M., Fitton, C. A., Steiner, M. F. C., McLay, J. S., Clark, D., King, A., Mackay, D. F., & Pell, J. P. (2017). Educational and health outcomes of children treated for attention-deficit/hyperactivity disorder. *JAMA Pediatrics*, **171**(7), e170691. https://doi.org/10.1001/jamapediatrics.2017.0691

Grzadzinski, R. L., Luyster, R., Spencer, A. G., & Lord, C. (2014). Attachment in young children with autism spectrum disorders: An examination of separation and Reunion behaviors with both mothers and fathers. *Autism*, **18**(2), 85–96. https://doi.org/10.1177/1362361312467235

Harada, S., Ito, H., Mochizuki, N., Tanaka, Y., Otake, S., Takayanagi, N., & Tsujii, M. (2014). Construct validity of the Japanese version of the strengths and difficulties questionnaire self-report form: Relations with depression, aggression, and parent-report form. *Japanese Journal of Child and Adolescent Psychiatry*, **53**, 343–351.

Herman, K. C., Lambert, S. F., Ialongo, N. S., & Ostrander, R. (2007). Academic pathways between attention problems and depressive symptoms among urban African American children. *Journal of Abnormal Child Psychology*, **35**(2), 265–274. https://doi.org/10.1007/s10802-006-9083-2

Hossain, M. M., Khan, N., Sultana, A., Ma, P., McKyer, E. L. J., Ahmed, H. U., & Purohit, N. (2020). Prevalence of comorbid psychiatric disorders among people with autism spectrum disorder: An umbrella review of systematic reviews and meta-analyses. *Psychiatry Research*, **287**, 112922. https://doi.org/10.1016/j.psychres.2020.112922

Hsiao, M. N., Tseng, W. L., Huang, H. Y., & Gau, S. S. F. (2013). Effects of autistic traits on social and school adjustment in children and adolescents: The moderating roles of age and gender. *Research in Developmental Disabilities*, **34**(1), 254–265. https://doi.org/10.1016/j.ridd.2012.08.001

Humphreys, K. L., Katz, S. J., Lee, S. S., Hammen, C., Brennan, P. A., & Najman, J. M. (2013). The association of ADHD and depression: Mediation by peer problems and parent–child difficulties in two complementary samples. *Journal of Abnormal Psychology*, **122**(3), 854–867. https://doi.org/10.1037/a0033895

Ichikawa, H., & Tanaka, Y. (2008). *ADHD rating scale for diagnosis and response: ADHD-RS [DSM compliant]*. Akashi Shoten.

Ito, H., Matsumoto, K., Takayanagi, N., Harada, A., Otake, S., Mochizuki, N., Nakajima, S., Noda, W., Tanaka, Y., & Tsujii, M. (2014a). Psychometric properties of the Japanese version of the autism Spectrum screening questionnaire (ASSQ): Development of a short form. *Japanese Journal of Psychology*, **85**(3), 304–312. https://doi.org/10.4992/jjpsy.85.13213

Ito, H., Tanaka, Y., Murayama, Y., Nakajima, T., Takayanagi, N., Noda, W., Mochizuki, N., Matsumoto, K., & Tsujii, M. (2014b). Development of the social maladjustment scale for elementary and junior high school students and verification of construct validity. *Clinical Psychiatry*, **56**, 699–708.

JASP Team. (2024). JASP (Version 0.18.1) [Computer software]. https://jasp-stats.org/

Jackson, Y., & Warren, J. S. (2000). Appraisal, social support, and life events: Predicting outcome behavior in school-age children. *Child Development*, **71**(5), 1441–1457. https://doi.org/10.1111/1467-8624.00238

Jones, W., Carr, K., & Klin, A. (2008). Absence of preferential looking to the eyes of approaching adults predicts level of social disability in 2-year-old toddlers with autism spectrum disorder. *Archives of General Psychiatry*, **65**(8), 946–954. https://doi.org/10.1001/archpsyc.65.8.946

Kamio, Y., Moriwaki, A., Takei, R., Inada, N., Inokuchi, E., Takahashi, H., & Nakahachi, T. (2013). Psychiatric issues of children and adults with autism spectrum disorders who remain undiagnosed. *Psychiatria et Neurologia Japonica*, **115**(6), 601–606.

Kasari, C., Locke, J., Gulsrud, A., & Rotheram-Fuller, E. (2011). Social networks and friendships at school: Comparing children with and without ASD. *Journal of Autism and Developmental Disorders*, **41**(5), 533–544. https://doi.org/10.1007/s10803-010-1076-x

Kawabata, Y., Tseng, W.-L., & Gau, S. S.-F. (2012). Symptoms of attention-deficit/hyperactivity disorder and social and school adjustment: The moderating roles of age and parenting. *Journal of Abnormal Child Psychology*, **40**(2), 177–188. https://doi.org/10.1007/s10802-011-9556-9

Kim, H. W., Cho, S. C., Kim, B. N., Kim, J. W., Shin, M. S., & Kim, Y. (2009). Perinatal and familial risk factors are associated with full syndrome and subthreshold attention-deficit hyperactivity disorder in a Korean community sample. *Psychiatry Investigation*, **6**(4), 278–285. https://doi.org/10.4306/pi.2009.6.4.278

Kofler, M. J., Rapport, M. D., & Alderson, R. M. (2008). Quantifying ADHD classroom inattentiveness, its moderators, and variability: A meta-analytic review. *Journal of Child Psychology and Psychiatry, and Allied Disciplines*, **49**(1), 59–69. https://doi.org/10.1111/j.1469-7610.2007.01809.x

Kraft, L., Ebner, C., Leo, K., & Lindenberg, K. (2023). Emotion regulation strategies and symptoms of depression, anxiety, aggression, and addiction in children and adolescents: A meta-analysis and systematic review. *Clinical Psychology: Science and Practice*, **30**(4), 485–502. https://doi.org/10.1037/cps0000156

Larson, K., Russ, S. A., Kahn, R. S., & Halfon, N. (2011). Patterns of comorbidity, functioning, and service use for US children with ADHD, 2007. *Pediatrics*, **127**(3), 462–470. https://doi.org/10.1542/peds.2010-0165

Lewinsohn, P. M., Shankman, S. A., Gau, J. M., & Klein, D. N. (2004). The prevalence and co-morbidity of subthreshold psychiatric conditions. *Psychological Medicine*, **34**(4), 613–622. https://doi.org/10.1017/S0033291703001466

Lundström, S., Chang, Z., Kerekes, N., Gumpert, C. H., Råstam, M., Gillberg, C., Lichtenstein, P., & Anckarsäter, H. (2011). Autistic-like traits and their association with mental health problems in two nationwide twin cohorts of children and adults. *Psychological Medicine*, **41**(11), 2423–2433. https://doi.org/10.1017/S0033291711000377

Mak, M. C. K., Yin, L., Li, M., Cheung, R. Y. H., & Oon, P. T. (2020). The relation between parenting stress and child behavior problems: Negative parenting styles as mediator. *Journal of Child and Family Studies*, **29**(11), 2993–3003. https://doi.org/10.1007/s10826-020-01785-3

Matsuoka, Y., Okada, R., Tani, I., Onishi, M., Nakajima, T., & Tsujii, M. (2011). Development of the parenting style scale: From developmental changes and relationships with ADHD tendencies. *Japanese Journal of Developmental Psychology*, **22**(2), 179–188. https://doi.org/10.11201/jjdp.22.179

Moriwaki, A., & Kamio, Y. (2013). Relationship between autistic behavioral characteristics and comorbid psychiatric symptoms in general children and students enrolled in regular elementary and junior high school classes in Japan. *Autism Spectrum Research*, **10**(1), 11–17.

Murata, T., Shimizu, A., Mori, Y., & Oshima, S. (1996). Childhood depression in school: A study using Birleson's depression self-rating scale for children. *Clinical Psychiatry*, **1**(2), 131–138.

Murayama, Y., Ito, H., Otake, S., Katagiri, M., Hamada, M., Nakajima, T., Uemiya, A., Nomura, K., Takayanagi, N., Myogan, K., & Tsujii, M. (2016). Cross-sectional effects of social support on mental health among elementary and junior high school students. *Japanese Journal of Developmental Psychology*, **27**(4), 395–407.

Nakajima, T., Ito, H., Myogan, K., Takayanagi, N., Murayama, Y., Hamada, M., Katori, M., & Tsujii, M. (2021). Relationships between autism spectrum disorder traits, recess play, and mental health: Verification among general

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elementary and junior high school students. *Japanese Journal of Developmental Psychology*, **32**, 233–244.

Namikawa, T., Tani, I., Wakita, T., Kumagai, R., Nakane, A., Noguchi, H., & Tsujii, M. (2011). Development of a short form of the Birleson depression self-rating scale for children (DSRS-C). *Clinical Psychiatry*, **53**(5), 489–496.

Noda, W., Okada, R., Tani, I., Ohnishi, M., Naoto, M., Nakajima, S., & Tsujii, M. (2013). Relationship among inattentive and hyperactive-impulsive behavior, aggression, and depression in Japanese elementary and junior high school students. *Shinrigaku Kenkyu*, **84**(2), 169–175. https://doi.org/10.4992/jjpsy.84.169

Orth, U., Meier, L. L., Bühler, J. L., Dapp, L. C., Krauss, S., Messerli, D., & Robins, R. W. (2024). Effect size guidelines for cross-lagged effects. *Psychological Methods*, **29**(2), 421–433. https://doi.org/10.1037/met0000499

Powell, V., Riglin, L., Hammerton, G., Eyre, O., Martin, J., Anney, R., Thapar, A., & Rice, F. (2020). What explains the link between childhood ADHD and adolescent depression? Investigating the role of peer relationships and academic attainment. *European Child and Adolescent Psychiatry*, **29**(11), 1581–1591. https://doi.org/10.1007/s00787-019-01463-w

Rai, D., Culpin, I., Heuvelman, H., Magnusson, C. M. K., Carpenter, P., Jones, H. J., Emond, A. M., Zammit, S., Golding, J., & Pearson, R. M. (2018). Association of autistic traits with depression from childhood to age 18 years. *JAMA Psychiatry*, **75**(8), 835–843. https://doi.org/10.1001/jamapsychiatry.2018.1323

Rapport, M. D., Kofler, M. J., Alderson, R. M., Timko, T. M., & Dupaul, G. J. (2009). Variability of attention processes in ADHD: Observations from the classroom. *Journal of Attention Disorders*, **12**(6), 563–573. https://doi.org/10.1177/1087054708322990

Rogers, M., & Tannock, R. (2018). Are classrooms meeting the basic psychological needs of children with ADHD symptoms? A self-determination theory perspective. *Journal of Attention Disorders*, **22**(14), 1354–1360. https://doi.org/10.1177/1087054713508926

Sakai, A., Yamasaki, K., Soga, S., Oashi, O., Shimai, S., & Otake, K. (2000). Development of an aggression questionnaire for elementary school children and examination of its reliability and validity. *Japanese Journal of School Health*, **42**(5), 423–433.

Simonoff, E., Pickles, A., Charman, T., Chandler, S., Loucas, T., & Baird, G. (2008). Psychiatric disorders in children with autism spectrum disorders: Prevalence, comorbidity, and associated factors in a population-derived sample. *Journal of the American Academy of Child and Adolescent Psychiatry*, **47**(8), 921–929. https://doi.org/10.1097/CHI.0b013e318179964f

Tanaka, Y., Ito, H., Murayama, Y., Noda, W., Nakajima, T., Hamada, M., Katagiri, M., Takayanagi, N., & Tsujii, M. (2015). Relationships between ASD and ADHD tendencies and bullying victimization and perpetration in nursery schools, elementary schools, and junior high schools. *Japanese Journal of Developmental Psychology*, **26**, 332–343.

Tanaka, Y., Ito, H., Takayanagi, N., Hrada, S., Noda, W., Otake, S., Nkajima, T., Mochizuki, N., & Tsujii, M. (2014). Examination of the effects of ASD and ADHD tendencies on peer relationship problems in elementary and junior high schools. *Clinical Psychiatry*, **56**, 501–510.

Uno, Y. (2020). Support and treatment in school age. *Japanese Journal of Psychiatric Treatment*, **35**, 172–175.

Whitehouse, A. J. O., Durkin, K., Jaquet, E., & Ziatas, K. (2009). Friendship, loneliness and depression in adolescents with Asperger's syndrome. *Journal of Adolescence*, **32**(2), 309–322. https://doi.org/10.1016/j.adolescence.2008.03.004

Williams, T., Connolly, J., Pepler, D., & Craig, W. (2005). Peer victimization, social support, and psychosocial adjustment of sexual minority adolescents. *Journal of Youth and Adolescence*, **34**(5), 471–482. https://doi.org/10.1007/s10964-005-7264-x

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